

Brutal Series B60 - Wastewater Story | Physical & Chemical Changes

| Large Numbers | Algebraic Expressions

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A treatment plant takes in 80,000 litres of sewage per hour. Screening removes 2% as large solids and grit removal then takes away 3% of the original volume. How many litres move on to the aeration tank?

[\[Wastewater Story\]](#)

- (A) 74000 L
- (B) 78400 L
- (C) 4000 L
- (D) 77600 L
- (E) 80000 L
- (F) 76048 L
- (G) 76000 L

2. Write 'six crore five lakh seventy thousand' as a single numeral. [\[Large Numbers\]](#)

- (A) 6057000
- (B) 600570000
- (C) 605700000
- (D) 6570000
- (E) 60057000
- (F) 65700000
- (G) 60570000

3. An iron gate of mass 12 kg rusts and gains 4% in mass because oxygen joins the iron. What is the mass of the rust (the added material), in grams? [\[Physical & Chemical Changes\]](#)

- (A) 12480 g
- (B) 240 g
- (C) 4800 g
- (D) 4.8 g
- (E) 480 g
- (F) 48 g

4. If $x = 3$ and $y = -2$, find the value of $2x^2 - 3xy + y^2$. [\[Algebraic Expressions\]](#)

- (A) -8
- (B) 22
- (C) 58
- (D) 40
- (E) 16
- (F) 4
- (G) 36

5. Sludge scraped from a settling tank is 92% water by mass. A 50 kg lump of this sludge is dried until only the solids remain. What mass of dry solids is left? [\[Wastewater Story\]](#)
- (A) 8 kg
 - (B) 12 kg
 - (C) 0.4 kg
 - (D) 4 kg
 - (E) 42 kg
 - (F) 46 kg
 - (G) 4.6 kg
6. The population of a town is 4,32,678. Rounded to the nearest ten thousand, what is it? [\[Large Numbers\]](#)
- (A) 433000
 - (B) 432680
 - (C) 440000
 - (D) 432000
 - (E) 430000
 - (F) 420000
7. A student lists six changes: melting wax, burning wax, cutting wax, dissolving sugar, souring of milk, and digesting food. How many of these are chemical changes? [\[Physical & Chemical Changes\]](#)
- (A) 6
 - (B) 5
 - (C) 4
 - (D) 1
 - (E) 0
 - (F) 3
8. Simplify $5a - 3b + 2a - (4a - b)$. [\[Algebraic Expressions\]](#)
- (A) $3a - 2b$
 - (B) $3a - 4b$
 - (C) $a - 2b$
 - (D) $11a - 2b$
 - (E) $3a + 2b$
 - (F) $3a - 3b$
 - (G) $7a - 4b$
9. A colony of 2,500 people each produce 150 litres of wastewater daily. The colony sewer can carry 300,000 litres a day. By how many litres does the daily wastewater exceed the sewer's capacity? [\[Wastewater Story\]](#)
- (A) 300000 L
 - (B) 75000 L
 - (C) 150000 L
 - (D) 675000 L
 - (E) 37500 L
 - (F) 25000 L

10. In the number 5,27,38,346, find the difference between the place values of the two digits 3. [\[Large Numbers\]](#)

- (A) 2997
- (B) 299970
- (C) 300300
- (D) 300000
- (E) 297000
- (F) 30000
- (G) 299700

11. A hot saturated solution weighs 250 g and is 40% dissolved salt. On cooling, 30 g of salt crystals separate out. What mass of salt is still dissolved in the solution? [\[Physical & Chemical Changes\]](#)

- (A) 30 g
- (B) 70 g
- (C) 40 g
- (D) 100 g
- (E) 130 g
- (F) 60 g
- (G) 150 g

12. A rectangle has length $(2x + 3)$ and breadth $(x - 1)$. Write its perimeter, given perimeter = $2(\text{length} + \text{breadth})$. [\[Algebraic Expressions\]](#)

- (A) $4x + 4$
- (B) $6x + 2$
- (C) $3x + 2$
- (D) $6x + 8$
- (E) $5x + 4$
- (F) $6x + 4$

13. In an aeration tank, bacteria reduce the organic waste by five-eighths in the first stage, then by half of what remains in the second stage. What fraction of the original organic waste is still left? [\[Wastewater Story\]](#)

- (A) $\frac{13}{16}$
- (B) $\frac{1}{8}$
- (C) $\frac{5}{16}$
- (D) $\frac{3}{16}$
- (E) $\frac{1}{16}$
- (F) $\frac{3}{8}$
- (G) $\frac{11}{16}$

14. Estimate the product $6,89,000 \times 31$ by first rounding each number to its greatest place. [\[Large Numbers\]](#)

- (A) 24000000
- (B) 18000000
- (C) 21000000
- (D) 2100000
- (E) 210000000
- (F) 21359000
- (G) 20000000

15. A metal sheet 50 cm by 40 cm is galvanised on both sides. The zinc coating needs 3 g of zinc per 100 square cm of surface. What total mass of zinc is used? [\[Physical & Chemical Changes\]](#)

- (A) 60 g
- (B) 240 g
- (C) 30 g
- (D) 6000 g
- (E) 120 g
- (F) 1200 g
- (G) 40 g

16. In the expression $7x^2 - x + 9$, what is the coefficient of x ? [\[Algebraic Expressions\]](#)

- (A) -1
- (B) 9
- (C) 1
- (D) -7
- (E) 7
- (F) -9

17. Chlorine is added at 4 milligrams per litre to disinfect treated water. How many grams of chlorine are needed to treat 250,000 litres? [\[Wastewater Story\]](#)

- (A) 10000 g
- (B) 1000 g
- (C) 1 g
- (D) 250 g
- (E) 62500 g
- (F) 100 g
- (G) 1000000 g

18. A factory makes 2,450 items every day. How many items does it make in a non-leap year of 365 days?

[\[Large Numbers\]](#)

- (A) 893250
- (B) 894250
- (C) 89425
- (D) 894200
- (E) 882000
- (F) 8942500
- (G) 901250

19. When magnesium burns, 24 g of magnesium combines with 16 g of oxygen to form 40 g of magnesium oxide. How many grams of oxide form when 9 g of magnesium burns completely? [\[Physical & Chemical Changes\]](#)

- (A) 9 g
- (B) 25 g
- (C) 6 g
- (D) 24 g
- (E) 40 g
- (F) 13 g
- (G) 15 g

20. Subtract $(3p^2 - 5p + 2)$ from $(7p^2 - 2p - 4)$. [\[Algebraic Expressions\]](#)

- (A) $4p^2 + 3p - 6$
- (B) $4p^2 + 3p + 6$
- (C) $10p^2 - 7p - 2$
- (D) $-4p^2 - 3p + 6$
- (E) $4p^2 - 7p - 2$
- (F) $4p^2 + 3p - 2$
- (G) $4p^2 + 7p - 6$

21. A home's used water is 40% from bathing, 25% from the kitchen, and the rest from washing clothes. If the kitchen water is 75 litres, how many litres come from washing clothes? [\[Wastewater Story\]](#)

- (A) 105 L
- (B) 35 L
- (C) 180 L
- (D) 75 L
- (E) 90 L
- (F) 120 L
- (G) 300 L

22. A publisher prints 3,75,000 copies of a book and sells 2,48,650 of them. How many copies remain, rounded to the nearest thousand? [\[Large Numbers\]](#)

- (A) 127000
- (B) 126350
- (C) 123350
- (D) 125000
- (E) 124000
- (F) 126000
- (G) 130000

23. From this list count the changes that can be reversed: freezing water, burning paper, folding paper, cooking an egg, and magnetising an iron bar. [\[Physical & Chemical Changes\]](#)

- (A) 1
- (B) 2
- (C) 6
- (D) 3
- (E) 0
- (F) 4
- (G) 5

24. A row of n squares made of matchsticks needs $(3n + 1)$ sticks. How many sticks are needed for 25 squares? [\[Algebraic Expressions\]](#)

- (A) 76
- (B) 73
- (C) 51
- (D) 26
- (E) 78
- (F) 100

25. An open drain is 30 m long and 0.4 m wide and holds stagnant water 0.05 m deep. What volume of water does it hold, in litres? (1 cubic metre = 1000 litres) [\[Wastewater Story\]](#)

- (A) 12 L
- (B) 600 L
- (C) 0.6 L
- (D) 6000 L
- (E) 6 L
- (F) 1200 L
- (G) 60 L

26. What is the successor of the largest seven-digit number? [\[Large Numbers\]](#)

- (A) 100000000
- (B) 11000000
- (C) 9999998
- (D) 1000000
- (E) 9999999
- (F) 10000000
- (G) 9999990

27. An iron rod loses 0.6 mm of thickness to rust each year. Starting at 9 mm thick, after how many years will it be reduced to half its original thickness? [\[Physical & Chemical Changes\]](#)

- (A) 8 years
- (B) 7.5 years
- (C) 15 years
- (D) 6 years
- (E) 4.5 years
- (F) 9 years

28. Find the value of $3(2x - 4) + 5$ when $x = 4$. [\[Algebraic Expressions\]](#)

- (A) 7
- (B) 5
- (C) 17
- (D) 29
- (E) 21
- (F) 12

29. Dried sludge is fed to a biogas unit. Each kilogram of dried sludge yields 0.45 cubic metres of biogas. How many cubic metres of biogas come from 1,200 kg of dried sludge? [\[Wastewater Story\]](#)

- (A) 54
- (B) 5400
- (C) 480
- (D) 600
- (E) 540
- (F) 270
- (G) 2666

30. Form an eight-digit number that has 5 in the crores place, 8 in the lakhs place, 2 in the thousands place, and 0 everywhere else. What is the number? [\[Large Numbers\]](#)

- (A) 508002000
- (B) 58002000
- (C) 50800200
- (D) 50082000
- (E) 5802000
- (F) 50820000
- (G) 50802000

31. When carbon burns, 12 g of carbon combines with 32 g of oxygen to make 44 g of carbon dioxide. What mass of carbon dioxide forms when 30 g of carbon burns fully? [\[Physical & Chemical Changes\]](#)

- (A) 80 g
- (B) 44 g
- (C) 132 g
- (D) 88 g
- (E) 110 g
- (F) 30 g

32. Combine all like terms in $4xy - 3x + 7yx + 2x^2 + 5x - xy$. [\[Algebraic Expressions\]](#)

- (A) $2x^2 + 10xy + 2x$
- (B) $10xy + 2x$
- (C) $2x^2 + 9xy + 2x$
- (D) $2x^2 + 11xy + 2x$
- (E) $2x^2 + xy + 2x$
- (F) $2x^2 + 10xy - 2x$

33. Treated water must have a BOD below 32 mg/L before release. A sample reads 80 mg/L. By what percentage must the BOD be reduced to just meet the limit? [\[Wastewater Story\]](#)

- (A) 150%
- (B) 20%
- (C) 68%
- (D) 60%
- (E) 48%
- (F) 40%

34. A rocket travels 1,08,000 km in one hour. How far does it travel in one full day of 24 hours? [\[Large Numbers\]](#)

- (A) 2598000
- (B) 2692000
- (C) 25920000
- (D) 2160000
- (E) 259200
- (F) 2592000

35. Salt dissolves at 36 g per 100 g of water at 50 C, but only 20 g per 100 g of water at 20 C. A solution made with 250 g of water is saturated at 50 C, then cooled to 20 C. How many grams of salt crystallise out?

[Physical & Chemical Changes]

- (A) 90 g
- (B) 140 g
- (C) 50 g
- (D) 36 g
- (E) 60 g
- (F) 40 g
- (G) 16 g

36. If $a = 2$, find the value of $3a^3 - 2a^2 + a$. [Algebraic Expressions]

- (A) 18
- (B) 14
- (C) 26
- (D) 30
- (E) 10
- (F) 22

37. A city of 600,000 people has proper sewers reaching only five-twelfths of them. How many people still lack proper sewerage? [Wastewater Story]

- (A) 350000
- (B) 300000
- (C) 360000
- (D) 420000
- (E) 250000
- (F) 50000
- (G) 150000

38. Estimate $8,12,560 - 2,97,310$ by first rounding each number to the nearest lakh. [Large Numbers]

- (A) 50000
- (B) 400000
- (C) 510000
- (D) 600000
- (E) 500000
- (F) 515250

39. The souring of milk is called a chemical change. Which statement gives the correct reason? [Physical &

Chemical Changes]

- (A) The milk merely changes its colour
- (B) No new substance is formed at all
- (C) Only the temperature of the milk changes
- (D) Only the state of the milk changes
- (E) It can be reversed by heating the milk
- (F) It is a very fast change
- (G) A new substance (acid) with different properties is formed

40. The phrase 'seven less than thrice a number n ' is written as an expression. What is its value when $n = 10$? [\[Algebraic Expressions\]](#)

- (A) 23
- (B) 13
- (C) 17
- (D) 37
- (E) 27
- (F) 21
- (G) 30

41. A septic tank is desludged every 3 years, and each desludging removes 1.8 cubic metres of sludge. Over 18 years, how many cubic metres of sludge are removed in total? [\[Wastewater Story\]](#)

- (A) 16.2
- (B) 5.4
- (C) 6
- (D) 1.8
- (E) 32.4
- (F) 10.8

42. Two numbers are 9,89,000 and 10,12,000. By how much is the larger greater than the smaller? [\[Large Numbers\]](#)

- (A) 2300
- (B) 32000
- (C) 13000
- (D) 230000
- (E) 23000
- (F) 21000
- (G) 123000

43. An iron piece of mass 200 g is galvanised and gains 5% in mass as zinc. Later it rusts and loses 2% of its new mass. What is its final mass? [\[Physical & Chemical Changes\]](#)

- (A) 200 g
- (B) 210 g
- (C) 215.8 g
- (D) 204 g
- (E) 205.8 g
- (F) 206 g
- (G) 196 g

44. Add the three expressions $(2x^2 + 3x - 5)$, $(-x^2 + x + 8)$ and $(3x^2 - 4x)$. [\[Algebraic Expressions\]](#)

- (A) $4x^2 + 13$
- (B) $6x^2 + 3$
- (C) $4x^2 + 3$
- (D) $4x^2 + 8x + 3$
- (E) $4x^2 - 3x + 3$
- (F) $4x^2 - 3$
- (G) $4x^2 + 3x + 3$

45. Wastewater flows through a house drain to the treatment inlet at 1.5 metres per second. The inlet is 270 metres away. How many seconds does the water take to reach it? [\[Wastewater Story\]](#)

- (A) 405 s
- (B) 180 s
- (C) 270 s
- (D) 18 s
- (E) 3 s
- (F) 1800 s

46. A water tanker holds 12,500 litres. A city stores 9,60,000 litres of water. How many full tankers can be filled from this store? [\[Large Numbers\]](#)

- (A) 80
- (B) 77
- (C) 70
- (D) 76
- (E) 75
- (F) 768
- (G) 76.8

47. 80 g of impure salt is dissolved and then recrystallised, giving 68 g of pure crystals; the rest stays in the solution as impurity. What percentage of the original sample was impurity? [\[Physical & Chemical Changes\]](#)

- (A) 32%
- (B) 85%
- (C) 12%
- (D) 18%
- (E) 17.6%
- (F) 20%
- (G) 15%

48. What is the numerical coefficient in the term $-5x^3y^2$? [\[Algebraic Expressions\]](#)

- (A) -5
- (B) 3
- (C) 5
- (D) -25
- (E) 6
- (F) -10

49. A treatment plant recovers reclaimed water equal to 75% of the sewage it takes in. To supply 30,000 litres of reclaimed water a day, how many litres of sewage must it take in? [\[Wastewater Story\]](#)

- (A) 10000 L
- (B) 37500 L
- (C) 22500 L
- (D) 40000 L
- (E) 30000 L
- (F) 45000 L
- (G) 52500 L

50. Write 'two crore three lakh nine thousand five' as a single numeral. [\[Large Numbers\]](#)

- (A) 20309005
- (B) 20300905
- (C) 2039005
- (D) 20309050
- (E) 20039005
- (F) 2309005

51. A candle of mass 60 g burns steadily at 2.5 g per hour. After 8 hours of burning, what mass of candle is left? [\[Physical & Chemical Changes\]](#)

- (A) 30 g
- (B) 45 g
- (C) 48 g
- (D) 20 g
- (E) 35 g
- (F) 50 g
- (G) 40 g

52. If $p = -1$, $q = 2$ and $r = -3$, find the value of $pq + qr + rp$. [\[Algebraic Expressions\]](#)

- (A) 1
- (B) -11
- (C) -7
- (D) 5
- (E) 11
- (F) -5

53. A factory's 600 litres of effluent is 90% contaminant. Clean water is added to dilute it to a 30% contaminant level. The contaminant amount stays the same. How many litres of clean water must be added? [\[Wastewater Story\]](#)

- (A) 360 L
- (B) 600 L
- (C) 1800 L
- (D) 1260 L
- (E) 540 L
- (F) 1200 L
- (G) 900 L

54. A company earns 3,650 rupees from each customer and has 1,20,000 customers. What is the total earning, expressed in crore rupees? [\[Large Numbers\]](#)

- (A) Rs 4.38 crore
- (B) Rs 43.8 lakh
- (C) Rs 43.8 thousand
- (D) Rs 43.8 crore
- (E) Rs 4.38 lakh
- (F) Rs 4380 crore
- (G) Rs 438 crore

55. From this list, count the chemical changes: setting of curd, ripening of a fruit, dissolving salt in water, the glowing of a bulb's filament, and the forming of clouds. [\[Physical & Chemical Changes\]](#)
- (A) 6
 - (B) 4
 - (C) 1
 - (D) 2
 - (E) 3
 - (F) 0
 - (G) 5
56. A triangle has sides $(x + 5)$, $(2x - 3)$ and $(3x + 1)$. If its perimeter is 39, find x . [\[Algebraic Expressions\]](#)
- (A) 7
 - (B) 36
 - (C) 6.5
 - (D) 12
 - (E) 5
 - (F) 6
57. A sludge heap of 800 kg loses 35% of its mass while composting. Then 20% of what remains is sold as manure. How many kilograms of compost are left unsold? [\[Wastewater Story\]](#)
- (A) 416 kg
 - (B) 280 kg
 - (C) 520 kg
 - (D) 440 kg
 - (E) 500 kg
 - (F) 364 kg
 - (G) 624 kg
58. A stadium seats 67,842 people. A headline rounds this to the nearest thousand. By how much does the rounded figure differ from the true seating? [\[Large Numbers\]](#)
- (A) 158
 - (B) 58
 - (C) 658
 - (D) 358
 - (E) 1000
 - (F) 842
59. When iron and sulphur react, 56 g of iron combines with 32 g of sulphur to form iron sulphide. What mass of sulphur is needed to react fully with 21 g of iron? [\[Physical & Chemical Changes\]](#)
- (A) 32 g
 - (B) 21 g
 - (C) 12 g
 - (D) 42 g
 - (E) 6 g
 - (F) 24 g
 - (G) 16 g

60. Subtract the sum of $(2a - 3b)$ and $(-a + 5b)$ from $(6a - 2b)$. [\[Algebraic Expressions\]](#)

- (A) $3a - 4b$
- (B) $5a$
- (C) $5a - 4b$
- (D) $7a$
- (E) $5a - 7b$
- (F) $5a + 4b$
- (G) $7a - 4b$

61. One pump empties 9,000 litres in 30 minutes; a second pump empties 12,000 litres in 30 minutes. Working together, how many minutes do they take to empty a 28,000-litre tank? [\[Wastewater Story\]](#)

- (A) 60 min
- (B) 30 min
- (C) 80 min
- (D) 20 min
- (E) 35 min
- (F) 40 min
- (G) 45 min

62. In the number 4,80,75,219, by how much does the place value of the 8 exceed the place value of the 7? [\[Large Numbers\]](#)

- (A) 7993000
- (B) 7930
- (C) 7999300
- (D) 7300000
- (E) 7930000
- (F) 8070000

63. A bridge is repainted to stop rust. It needs 3 coats, and each coat covers 250 square metres per 20 litres of paint. For a 6,000 square metre surface, how many litres of paint are needed for all 3 coats? [\[Physical & Chemical Changes\]](#)

- (A) 160 L
- (B) 480 L
- (C) 1200 L
- (D) 720 L
- (E) 360 L
- (F) 1440 L
- (G) 960 L

64. If $x = \frac{1}{2}$, find the value of $8x^2 + 4x - 1$. [\[Algebraic Expressions\]](#)

- (A) 4
- (B) 3
- (C) 5
- (D) 2
- (E) 1
- (F) 0
- (G) 6

65. Treatment drops a sample's BOD from 240 mg/L to 60 mg/L. A polishing step then cuts the leftover to one-third of 60. What is the ratio of the final released BOD to the original BOD, in simplest form? [\[Wastewater Story\]](#)

- (A) 1:4
- (B) 1:3
- (C) 1:20
- (D) 1:36
- (E) 1:12
- (F) 1:8
- (G) 1:6

66. A train carries 1,250 passengers per trip and makes 18 trips a day. How many passengers does it carry in 30 days? [\[Large Numbers\]](#)

- (A) 67500
- (B) 562500
- (C) 6750000
- (D) 675000
- (E) 675
- (F) 787500
- (G) 22500

67. Which statement is TRUE of a physical change? [\[Physical & Chemical Changes\]](#)

- (A) It always changes the colour
- (B) It always forms a solid deposit
- (C) It always gives off a gas
- (D) It always releases a lot of heat
- (E) No new substance forms and it is often reversible
- (F) A new substance always forms

68. The terms of a pattern follow $n\text{th term} = 4n - 3$, giving 1, 5, 9, 13, ... Which term of this pattern is equal to 77? [\[Algebraic Expressions\]](#)

- (A) 17
- (B) 18
- (C) 21
- (D) 20
- (E) 77
- (F) 19
- (G) 80

69. Treating 1,000 litres of sewage costs 6 rupees. A town treats 4.5 lakh litres each day. What is the daily treatment cost? [\[Wastewater Story\]](#)

- (A) Rs 2700
- (B) Rs 750000
- (C) Rs 27
- (D) Rs 2700000
- (E) Rs 27000
- (F) Rs 4500
- (G) Rs 270

70. A number is written in expanded form as $7 \times 10,00,000 + 4 \times 10,000 + 9 \times 100 + 6$. What is the number? [\[Large Numbers\]](#)

- (A) 7040906
- (B) 7040096
- (C) 704906
- (D) 7400906
- (E) 7400096
- (F) 7049006
- (G) 70040906

71. Mixing baking soda with vinegar releases carbon dioxide gas, a chemical change. If 5 g of baking soda releases 2.6 litres of gas, how many litres are released by 12.5 g of baking soda? [\[Physical & Chemical Changes\]](#)

- (A) 3.25 L
- (B) 5.2 L
- (C) 6.5 L
- (D) 26 L
- (E) 1.04 L
- (F) 7.8 L

72. Which of the following expressions is a binomial? [\[Algebraic Expressions\]](#)

- (A) 9
- (B) $4abc$
- (C) $2x + 3y - 1$
- (D) $x^2 + 2x + 1$
- (E) $7xyz$
- (F) $3x - 5$

73. A sewer pipe is laid so it falls 4 cm for every 5 m of length. Over a 1.25 km line, what is the total fall, in metres? [\[Wastewater Story\]](#)

- (A) 100 m
- (B) 1 m
- (C) 50 m
- (D) 20 m
- (E) 10 m
- (F) 1000 m

74. Estimate the quotient $5,94,000 / 29$ by rounding each number suitably first. [\[Large Numbers\]](#)

- (A) 22000
- (B) 200000
- (C) 2000
- (D) 20484
- (E) 20000
- (F) 200
- (G) 18000

- 75.** Under the same conditions, a saturated solution gives 9 g of crystals from every 60 g of solution. How many grams of solution are needed to obtain 60 g of crystals? [\[Physical & Chemical Changes\]](#)
- (A) 400 g
 - (B) 540 g
 - (C) 90 g
 - (D) 360 g
 - (E) 40 g
 - (F) 450 g
 - (G) 600 g
- 76.** A taxi charges Rs $(40 + 12k)$ for a ride of k kilometres. What is the fare for a 15 km ride? [\[Algebraic Expressions\]](#)
- (A) Rs 600
 - (B) Rs 180
 - (C) Rs 200
 - (D) Rs 220
 - (E) Rs 192
 - (F) Rs 240
 - (G) Rs 52
- 77.** Two villages share one plant. Village A sends 3 parts and Village B sends 5 parts of a total daily flow of 4,800 litres. How many litres does Village A send? [\[Wastewater Story\]](#)
- (A) 600 L
 - (B) 1800 L
 - (C) 1600 L
 - (D) 2400 L
 - (E) 3000 L
 - (F) 2000 L
 - (G) 1200 L
- 78.** A donor gives 2 crore rupees to be shared equally among 800 students. How much does each student receive? [\[Large Numbers\]](#)
- (A) Rs 2500000
 - (B) Rs 250
 - (C) Rs 2500
 - (D) Rs 16000
 - (E) Rs 20000
 - (F) Rs 25000
 - (G) Rs 250000
- 79.** When methane burns, 16 g of methane reacts with 64 g of oxygen. How many grams of oxygen are needed to burn 4 g of methane completely? [\[Physical & Chemical Changes\]](#)
- (A) 8 g
 - (B) 48 g
 - (C) 64 g
 - (D) 16 g
 - (E) 32 g
 - (F) 4 g
 - (G) 80 g

80. Simplify $3(x - 2) - 2(x - 5)$ and then find its value when $x = 4$. [\[Algebraic Expressions\]](#)

- (A) 2
- (B) 16
- (C) 8
- (D) 12
- (E) 0
- (F) 6
- (G) 4

81. Along a 2.4 km sewer, manholes are placed every 30 m, with one at each end. How many manholes are there in all? [\[Wastewater Story\]](#)

- (A) 160
- (B) 8
- (C) 80
- (D) 81
- (E) 82
- (F) 800
- (G) 79

82. Using the digits 0, 4, 7, 2, 9 and 5 each exactly once, what is the smallest six-digit number you can form? [\[Large Numbers\]](#)

- (A) 207459
- (B) 204579
- (C) 205479
- (D) 204597
- (E) 240579
- (F) 204759

83. From this list, count the irreversible (chemical) changes: melting of ice, rusting of iron, burning of a candle, evaporation of petrol, digestion of food, breaking of glass, and cooking of rice. [\[Physical & Chemical Changes\]](#)

- (A) 4
- (B) 5
- (C) 6
- (D) 7
- (E) 2
- (F) 1

84. In the expression $9 - 2x + 7x - 5x^2 + 3x$, what is the sum of the coefficients of all the x terms? [\[Algebraic Expressions\]](#)

- (A) 5
- (B) -2
- (C) 10
- (D) 3
- (E) 12
- (F) 7
- (G) 8

- 85.** A school uses 1,500 litres of water a day; 60% of it becomes greywater, and 70% of that greywater is reused for gardening. How many litres are reused? [\[Wastewater Story\]](#)
- (A) 270 L
 - (B) 1050 L
 - (C) 630 L
 - (D) 450 L
 - (E) 900 L
 - (F) 700 L
- 86.** A satellite completes one orbit of Earth every 96 minutes. How many complete orbits does it make in 7 days? [\[Large Numbers\]](#)
- (A) 70
 - (B) 105
 - (C) 96
 - (D) 210
 - (E) 10080
 - (F) 104
 - (G) 1050
- 87.** 500 g of a solution is 28% salt. All the water is evaporated to crystallise the salt, but 6% of the salt is lost as spray. What mass of salt crystals is recovered? [\[Physical & Chemical Changes\]](#)
- (A) 8.4 g
 - (B) 132 g
 - (C) 125 g
 - (D) 130 g
 - (E) 148.4 g
 - (F) 131.6 g
 - (G) 140 g
- 88.** A square has side $(2x + 1)$. Using perimeter = 4 x side, find the perimeter when $x = 3$. [\[Algebraic Expressions\]](#)
- (A) 7
 - (B) 21
 - (C) 24
 - (D) 49
 - (E) 32
 - (F) 28
 - (G) 14
- 89.** Untreated sewage carries 8,000,000 bacteria per litre. Chlorination kills 99.5% of them. How many bacteria survive in each litre? [\[Wastewater Story\]](#)
- (A) 7960000
 - (B) 79600
 - (C) 400000
 - (D) 80000
 - (E) 40000
 - (F) 4000
 - (G) 8000

90. In the number 6,07,48,512, what value does the digit 7 stand for? [\[Large Numbers\]](#)

- (A) 7000000
- (B) 70000
- (C) 700000
- (D) 70000000
- (E) 7000
- (F) 7
- (G) 70

91. Rust eats into an iron pipe from the outside at 0.2 mm of radius per month. After 2 years, how many millimetres of radius have been lost? [\[Physical & Chemical Changes\]](#)

- (A) 0.48 mm
- (B) 1.2 mm
- (C) 5 mm
- (D) 9.6 mm
- (E) 2.4 mm
- (F) 4.8 mm
- (G) 48 mm

92. From $5x^2 - 3x + 2$ subtract $2x^2 + x - 7$, then find the value of the result when $x = -1$. [\[Algebraic Expressions\]](#)

- (A) 22
- (B) 16
- (C) 2
- (D) 10
- (E) 8
- (F) 6
- (G) 12

93. A drain carries 18 cubic metres of water in one hour. Express this flow rate in litres per minute. (1 cubic metre = 1000 litres) [\[Wastewater Story\]](#)

- (A) 300 L/min
- (B) 30 L/min
- (C) 180 L/min
- (D) 18000 L/min
- (E) 1080000 L/min
- (F) 1800 L/min
- (G) 3000 L/min

94. The exact product of 405 and 980 is compared with the estimate found by using 400 and 1000. By how much does the estimate exceed the exact product? [\[Large Numbers\]](#)

- (A) 400000
- (B) 2100
- (C) 4100
- (D) 396900
- (E) 3100
- (F) 3900
- (G) 31000

95. Burning 1 kg of wood leaves 50 g of ash. How many kilograms of ash are left after burning 24 kg of wood? [\[Physical & Chemical Changes\]](#)

- (A) 120 kg
- (B) 0.12 kg
- (C) 2.4 kg
- (D) 1.2 kg
- (E) 12 kg
- (F) 0.6 kg
- (G) 24 kg

96. In a dot pattern, figure number n contains $(n^2 + n)$ dots. How many dots are in figure 5? [\[Algebraic Expressions\]](#)

- (A) 36
- (B) 25
- (C) 31
- (D) 11
- (E) 20
- (F) 35
- (G) 30

97. A sanitation budget of 48 lakh rupees gives five-eighths to laying pipelines and the rest to a treatment plant. How much, in lakh rupees, goes to the treatment plant? [\[Wastewater Story\]](#)

- (A) Rs 6 lakh
- (B) Rs 24 lakh
- (C) Rs 30 lakh
- (D) Rs 8 lakh
- (E) Rs 18 lakh
- (F) Rs 42 lakh

98. A printing press prints 9,000 sheets per hour and runs for 7.5 hours a day. How many sheets does it print in 12 days? [\[Large Numbers\]](#)

- (A) 67500
- (B) 8100000
- (C) 8100
- (D) 810000
- (E) 81000
- (F) 945000
- (G) 675000

99. Freezing water is a physical change: the mass stays the same but the volume grows. If 2.5 litres of water freeze and expand by 9% in volume, what is the volume of the ice formed? [\[Physical & Chemical Changes\]](#)

- (A) 2.225 L
- (B) 2.5 L
- (C) 2.95 L
- (D) 2.725 L
- (E) 0.225 L
- (F) 2.75 L

100. The sum of three consecutive integers, with the first equal to n , is $3n + 3$. If this sum is 72, what is the middle integer? [\[Algebraic Expressions\]](#)

- (A) 72
- (B) 25
- (C) 26
- (D) 21
- (E) 23
- (F) 22
- (G) 24